FCBsP: Fixed-Constellation Belief-selective Propagation Detection for MIMO Turbo Receivers

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***Abstract*—The belief-selective propagation (BsP) algorithm has recently emerged as a promising approach for massive MIMO detection. However, when applied in MIMO turbo receivers, known for their superior performance compared to separated detection and decoding (SDD) receivers, the BsP-based receiver suffers from significant performance degradation and high processing latency. To overcome these limitations, this paper proposes a fixed-constellation BsP (FCBsP) detector tailored for MIMO turbo receivers. By building *fixed configuration sets* and utilizing the *approximate multi-user interference* for message updates, the proposed FCBsP detector achieves a better trade-off between error performance and computational complexity compared to the BsP. Furthermore, two unexplored features: *information compensation* and *decoding-first mechanism* are proposed to fine-tune the exchanged information and lower the processing latency of the FCBsP-based turbo receiver. Numerical results demonstrate that the proposed FCBsP-based turbo receiver earns about 0.7 and 1.8 dB performance gains over the BsP-based turbo receiver at BLER=10^{-3} in an LDPC-coded 32×12 64-QAM MIMO system under Rayleigh and practical channels, respectively.**

***Index Terms*—Multiple-input multiple-output (MIMO), belief propagation (BP), low-density parity-check (LDPC), MIMO detection, iterative detection and decoding (IDD).**

I. Introduction

RAPID advancements in wireless communication technology have revolutionized modern society, catalyzing remarkable progress in living standards globally. Multiple-input multiple-output (MIMO), a "cornerstone" of modern wireless communication systems, is advancing toward massive antenna deployments to meet the stringent requirements of the forthcoming 6G applications [1, 2]. Therefore, signal detection in large-scale MIMO systems becomes a challenging issue [3, 4]. Specifically, the well-known *maximum a posteriori* (MAP) detector achieves optimal performance but incurs prohibitive complexity as the antenna scale increases. In contrast, the linear minimum mean square error (LMMSE) detector [5], known for its low computational complexity and high hardware feasibility [6], is widely adopted in massive MIMO systems. However, the LMMSE detector experiences significant performance degradation in challenging MIMO scenarios, e.g., enhanced mobile broadband (eMBB) and ultra-reliable and low-latency communications (URLLC) [7].

Iterative detectors relying on Bayesian message passing, such as expectation propagation (EP) detectors [8]–[10], approximate message passing (AMP) detectors [11]–[13], and belief propagation (BP) detectors [14]–[16], have emerged as promising techniques due to their effective trade-offs between error performance and computational complexity. Specifically, the EP detector introduces moment matching to approximate the posterior messages as Gaussian distributions, achieving near-optimal bit error rate (BER) performance in various MIMO scenarios. However, the EP detector requires matrix inversion operations during each iteration, resulting in relatively low hardware efficiency for massive MIMO systems. The AMP detector serves as an alternative Bayesian-optimal detector for MIMO systems satisfying the large-system limit. Several variants of the AMP have been developed to improve performance robustness in ill-conditioned channels, such as orthogonal AMP (OAMP) [12] and memory AMP (MAMP) [13]. However, the AMP-based detectors often suffer from low convergence, and their performance deteriorates when the MIMO dimension is not sufficiently large. In contrast, the BP detector circumvents the matrix inversion operations by updating messages based on the factor graph, offering near-optimal BER performance and enhanced hardware feasibility. Among the low-complexity variants [15, 17] of BP detector, the real-domain Gaussian approximation of interference BP (RD-GAI-BP) is widely adopted [18]. However, it suffers from a performance error floor with relatively high computational costs in large-scale MIMO scenarios. Recently, belief-selective propagation (BsP) detection has been proposed as a promising alternative [19]–[21]. By selecting incoming messages with high reliability for updates, BsP detection effectively mitigates the error floor issue and reduces computational complexity. Moreover, an implementation-friendly version of the BsP, named approximate BsP (aBsP), along with its corresponding ASIC implementation, was proposed in [20]. Despite its performance advantages and implementation benefits for massive MIMO detection, the BsP detector is not well-suited for direct application in coded MIMO systems, particularly in turbo MIMO systems. On one hand, the BsP detector leads to performance degradation in MIMO turbo receivers due to its over-truncated constellations. On the other hand, it still suffers from non-negligible processing latency, which is crucial for the performance of MIMO turbo receivers.

In coded MIMO systems, receivers are typically categorized into separated detection and decoding (SDD) receivers and

This work was supported in part by NSFC under Grants 62331009, in part by the Jiangsu Provincial NSF under BG2024004 and BM2023016, and in part by the Fundamental Research Funds for the Central Universities.

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iterative detection and decoding (IDD) receivers [22]. In particular, SDD receivers adopt a one-way information flow, where the detector outputs soft or hard symbol estimates to the decoder, but no feedback is provided from the decoder to the detector. The SDD receivers enjoy low-latency merits, making them widely adopted in practical systems. In contrast, IDD receivers, also known as MIMO turbo receivers, enable iterative exchange of soft information between the detector and decoder. The IDD mechanism iteratively refines the soft information, contributing to superior error performance compared to SDD receivers. Therefore, the turbo architecture has been increasingly applied to various communication scenarios. For instance, several studies in the field of grant-free random access for 5G massive machine-type communications (mMTC) have leveraged the turbo architecture to improve communication reliability [23, 24]. However, IDD receivers face challenges of high processing latency and low throughput. To address these problems, there exist several studies focusing on the optimization of turbo architecture-based EP detectors [25]–[27] and AMP detectors [24, 28] to achieve lower computational complexity and processing latency. However, to the author's best knowledge, existing BP-based IDD receivers [29, 30] typically incorporate the BP detector directly within the IDD framework, without specifically optimizing it for the IDD structure. As a result, the complexity of BP-based IDD receivers is prohibitive, limiting their practical applicability in MIMO systems.

To address these challenges, a fixed-constellation BsP detector (FCBsP) tailored for MIMO turbo receivers is proposed, which can not only reduce the computational complexity but also achieve receiver performance gains compared with the BsP detector. Additionally, we introduce the information compensation strategy to mitigate potential information loss during data transmission between the detector and decoder in the IDD receiver. Furthermore, the mechanism of the IDD receiver is optimized, enabling adaptive reduction of overall latency based on varying channel conditions. The main contributions of this paper are as follows.

- **Proposing a fixed-constellation BsP (FCBsP) detector:** Facing the customized requirements for MIMO turbo receivers, the FCBsP detector is introduced in this paper. Specifically, the proposed FCBsP detector incorporates two unexplored features: *fixed configuration set* and *approximate multi-user interference*, resulting in a better trade-off between error performance and computational complexity compared to the BsP detector. Numerical results show that the proposed FCBsP detector achieves nearly 0.5 dB performance gains over the BsP detector with 58% complexity reduction in the large-scale (128×64) MIMO system with 256-QAM modulation under the Rayleigh channel.
- **Designing an FCBsP-tailored MIMO turbo receiver:** Toward the low-latency and high-reliable demands, an FCBsP-tailored MIMO turbo receiver is designed in this paper. More specifically, first, an *information compensation* strategy is introduced to address the missing counter-hypotheses issue [31], mitigating the performance loss of the FCBsP turbo receiver. Second, a *decoding-first* mechanism is proposed for the MIMO turbo receiver, sufficiently reducing the processing latency without performance penalization. Numerical results demonstrate that in a 5G $(1440, 1152)$[1] LDPC-coded 32×12 64-QAM MIMO system, the proposed decoding-first mechanism can save up to 93.91% processing latency, and the information compensation strategy can contribute 1.0 dB gains to the FCBsP turbo receiver. Compared with the BsP turbo receiver, the proposed FCBsP turbo receiver achieves 0.7 dB and 1.8 dB performance gains over the BsP turbo receiver under the Rayleigh and practical channels, respectively.
- **Achieving faster convergence and lower complexity:** The convergence feature of the proposed FCBsP detector is analyzed to demonstrate its performance effectiveness in both uncoded and coded MIMO scenarios. The results demonstrate that the proposed FCBsP turbo receiver achieves faster convergence speed compared to other turbo receivers. In addition, complexity analysis results with Fixed-Point Operations (FxPOs) [20] are also provided. Compared with the BsP detector, the proposed FCBsP detector delivers up to 58% complexity reduction in an 8×4 MIMO scenario with 256-QAM modulation.

The remainder of this paper is organized as follows. Section II introduces the background of the MIMO turbo system model and the BsP detector. Section III details the techniques of the proposed FCBsP detector and its corresponding turbo receiver. Section IV shows the convergence and complexity analysis of the proposed FCBsP. Section V offers the numerical results. Section VI concludes this paper.

Notations: We utilize boldface lowercase letter $\mathbf{a}$ and uppercase letter $\mathbf{A}$ to indicate a column vector and a matrix, respectively. $(\cdot)^T$ denotes the transpose operator. $\mathbf{I}_{N_t}$ is an identity matrix of size $N_t \times N_t$. diag$\{\mathbf{A}\}_j$ represents the entry positioned at the j-th column and the j-th row in $\mathbf{A}$. We apply $\mathcal{CN}(\mu, \sigma^2)$ to denote the complex-valued Gaussian distribution with μ mean and variance σ^2.

II. Preliminaries

A. System Model

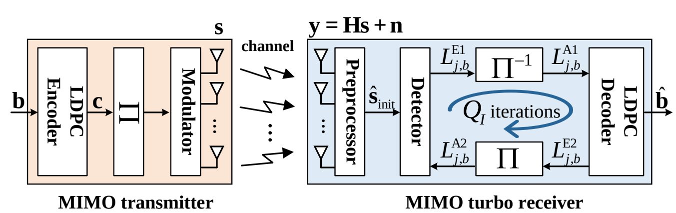


Fig. 1: System diagram for a MIMO turbo receiver.

Without loss of generality, we consider an uplink LDPC-coded MIMO system with a turbo receiver as illustrated in Fig. 1. The MIMO system consists of N_t transmit antennas and N_r receive antennas. On the transmitter side, an information

[1]Let (N, K) denote an LDPC code with a code length of N and K information bits.

bit vector $\mathbf{b} \in \{0,1\}^{K\times 1}$ with length K is fed into the LDPC encoder, which outputs an encoded codeword $\mathbf{c}$ of length N. Thus, the code rate is given by $R_c = K/N$. Then, the codeword is interleaved by the interleaver Π and subsequently input into a modulator with modulation order M. In the modulation process, the interleaved bits are grouped into several M-bit sequences, which are mapped to unit-average-power symbols using $|\tilde{\mathcal{Q}}|$-quadrature amplitude modulation (QAM). The constellation $\tilde{\mathcal{Q}}$ has the cardinality of $|\tilde{\mathcal{Q}}| = 2^M$. These modulated symbols are divided into several groups of length N_t. For simplification and without loss of generality, we focus on a single group transmitted through the MIMO channel, i.e., $\tilde{\mathbf{s}} = [\tilde{s}_1, ..., \tilde{s}_{N_t}]^T \in \tilde{\mathcal{Q}}^{N_t\times 1}$. On the receiver side, the received symbol vector $\tilde{\mathbf{y}} \in \mathbb{C}^{N_r\times 1}$ can be expressed as

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}\tilde{\mathbf{s}} + \tilde{\mathbf{n}}, \tag{1}$$

where $\tilde{\mathbf{H}} \in \mathbb{C}^{N_r\times N_t}$ represents a complex channel matrix and $\tilde{\mathbf{n}} \in \mathbb{C}^{N_r\times 1}$ is a complex additive white Gaussian noise (AWGN) vector with zero-mean and variance σ_n^2, i.e., $\tilde{n}_i \sim \mathcal{CN}(0, \sigma_n^2)$.

To simplify the complex-valued system model in Eq. (1), we convert it into an equivalent real-valued model by separating the real and imaginary components. Specifically, we define $\mathbf{s} = \begin{bmatrix} \mathrm{Re}(\tilde{\mathbf{s}}) \\ \mathrm{Im}(\tilde{\mathbf{s}}) \end{bmatrix} \in \mathcal{Q}^{2N_t\times 1}$, $\mathbf{n} = \begin{bmatrix} \mathrm{Re}(\tilde{\mathbf{n}}) \\ \mathrm{Im}(\tilde{\mathbf{n}}) \end{bmatrix} \in \mathbb{R}^{2N_r\times 1}$, and $\mathbf{y} = \begin{bmatrix} \mathrm{Re}(\tilde{\mathbf{y}}) \\ \mathrm{Im}(\tilde{\mathbf{y}}) \end{bmatrix} \in \mathbb{R}^{2N_r\times 1}$. Note that $\mathcal{Q}$ is the real-version constellation set of $\tilde{\mathcal{Q}}$, with its size given by $|\mathcal{Q}| = \sqrt{|\tilde{\mathcal{Q}}|}$. The real-valued channel matrix $\mathbf{H}$ is acquired as

$$\mathbf{H} = \begin{bmatrix} \mathrm{Re}(\tilde{\mathbf{H}}) & -\mathrm{Im}(\tilde{\mathbf{H}}) \\ \mathrm{Im}(\tilde{\mathbf{H}}) & \mathrm{Re}(\tilde{\mathbf{H}}) \end{bmatrix} \in \mathbb{R}^{2N_r\times 2N_t}. \tag{2}$$

Therefore, the equivalent system model is given as

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{n}. \tag{3}$$

This paper assumes the channel matrix is perfectly known at the receiver. After receiving the symbol vector $\mathbf{y}$, the turbo receiver inputs it into the preprocessor. The preprocessor will generate the initial information of transmitted (TX) symbols for the detector. Typically, the LMMSE algorithm is employed [19, 32] to provide initial symbol estimates, which are given by

$$\hat{\mathbf{s}}_{\text{init}} = (\mathbf{H}^T\mathbf{H} + \frac{\sigma_n^2}{2}\mathbf{I}_{2N_t})^{-1}\mathbf{H}^T\mathbf{y}. \tag{4}$$

Then, the turbo receiver iteratively exchanges the extrinsic information between the detector and the LDPC decoder to improve the reliability of the soft information for each coded bit. Specifically, let $L_{j,b}^{E1}$ and $L_{j,b}^{E2}$ denote the extrinsic output bit LLR of the detector and decoder, respectively, for the b-th bit of the j-th symbol. Correspondingly, $L_{j,b}^{A1}$ and $L_{j,b}^{A2}$ represent the *a priori* input LLR value to the decoder and detector, respectively. To begin with, the detector produces the symbol LLR $\gamma_j(k)$ for the j-th TX symbol corresponding to μ_k, the k-th real constellation symbol in $\mathcal{Q}$. The symbol LLR is defined as

$$\gamma_j(k) = \log\frac{p(s_j = \mu_k)}{p(s_j = \mu_1)}, \tag{5}$$

where μ_1 is selected as the reference symbol. Next, the symbol LLR is mapped into the bit LLR $L_{j,b}^{\text{E1}}$ as

$$\begin{aligned} L_{j,b}^{\text{E1}} &= \log\left(\frac{\sum_{\mu_k\in\mathcal{Q}_b^1}\exp(\gamma_j(k))}{\sum_{\mu_k\in\mathcal{Q}_b^0}\exp(\gamma_j(k))}\right) \\ &\overset{\max-\log}{\approx} \max_{\mu_k\in\mathcal{Q}_b^1}\gamma_j(k) - \max_{\mu_k\in\mathcal{Q}_b^0}\gamma_j(k), \end{aligned} \tag{6}$$

where $\mathcal{Q}_b^1$ and $\mathcal{Q}_b^0$ represent the constellation sets corresponding to the b-th bit being 1 and 0, respectively. *Max-log* approximation operation is employed for complexity reduction, which can be expressed as $\log\left(\sum_k e^{x_k}\right) \approx \max(x_k)$. Subsequently, the extrinsic output $(L_{j,b}^{\text{E1}})$ is sent to the deinterleaver to produce the *a priori* input $(L_{j,b}^{\text{A1}})$ for the LDPC decoder. In the backward information transmission, the LDPC decoder generates its extrinsic LLRs $(L_{j,b}^{\text{E2}})$, which is re-interleaved to yield the *a priori* LLR $(L_{j,b}^{A2})$ for the detector to begin the next receiver iteration. Then, the bit LLRs are remapped to the symbol LLR as

$$\gamma_j(k) = \sum_{b=1}^{M/2} L_{j,b}^{\text{A2}}\cdot\mu_k(b), \tag{7}$$

where $\mu_k(b) \in \{0,1\}$ denotes the b-th bit for μ_k. Finally, the turbo receiver will output its estimated bit vector $\hat{\mathbf{b}}$ until the output bits of the decoder match the early stopping criterion or the receiver reaches its maximum iteration limit Q_I.

B. Belief-selective Propagation Detector

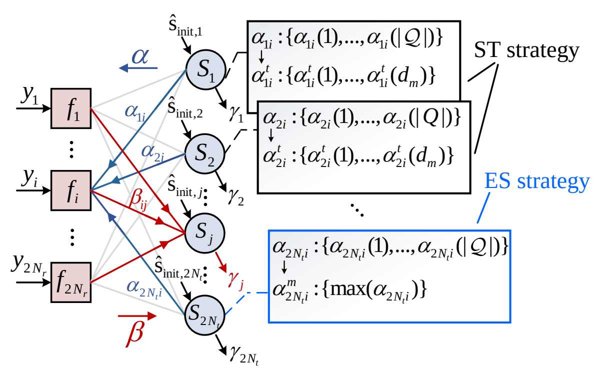


Fig. 2: Factor graph of the BsP detector.

The BsP detector is activated when receiving the initial results of TX symbols from the preprocessor, which are then utilized to compute the initial γ messages as

$$\gamma_j(k) = \frac{\|\mu_1 - \hat{s}_{\text{init},j}\|^2 - \|\mu_k - \hat{s}_{\text{init},j}\|^2}{2\sigma_{\text{init},j}^2}, \tag{8}$$

where $\hat{s}_{\text{init},j}$ represents the j-th element in $\hat{\mathbf{s}}_{\text{init}}$, and $\sigma_{\text{init},j}^2$ is the j-th diagonal element in the covariance matrix $\mathbf{K}$. The covariance matrix is given as

$$\mathbf{K} = (\mathbf{H}^T\mathbf{H} + \frac{\sigma_n^2}{2}\mathbf{I}_{2N_t})^{-1}. \tag{9}$$

Then, the BsP detector iteratively updates soft messages based on the factor graph. As illustrated in Fig. 2, the

$$\beta_{ij}(k) = \max_{\substack{\mathbf{s}:s_j=\mu_k,\\ \mathbf{s}_{\mathbf{k}/j}\in\mathcal{B}(d_m,d_f)}} \left\{-\frac{1}{2\sigma_n^2}\|y_i-\mathbf{h}_i\mathbf{s}\|^2+\sum_{t=1,t\neq j}^{2N_t}\alpha_{ti}\right\} - \max_{\substack{\mathbf{s}:s_j=\mu_1,\\ \mathbf{s}_{\mathbf{k}/j}\in\mathcal{B}(d_m,d_f)}} \left\{-\frac{1}{2\sigma_n^2}\|y_i-\mathbf{h}_i\mathbf{s}\|^2+\sum_{t=1,t\neq j}^{2N_t}\alpha_{ti}\right\} \quad (12)$$

factor graph of the BsP detector includes $2N_r$ factor nodes (FNs) and $2N_t$ symbol nodes (SNs), which are denoted as f_i and S_j, respectively. Soft messages sent from f_i to S_j are represented by β_{ij} messages, and those sent from S_j to f_i are denoted by α_{ji} messages. To update messages in relatively low complexity, the BsP detector utilizes two effective strategies: *symbol-based truncation* (ST) and *edge-based simplification* (ES). Specifically, to update the message β_{ij} with $2N_t-1$ incoming edges, the BsP detector arbitrarily selects d_f-1 edges as *chosen edges*, while the remained $2N_t-d_f$ edges are treated as *simplified edges*. As shown in Fig. 2, the α messages corresponding to chosen edges are truncated by utilizing the ST strategy, reserving the largest d_m symbol LLRs (denoted as α_{ji}^t). To achieve this, we define a permutation π that sorts the $\alpha_{ji}(k)$ in descending order of reliability, i.e., $\alpha_{ji}(\pi(1)) > \alpha_{ji}(\pi(2)) > \cdots > \alpha_{ji}(\pi(d_m))$. The sorted LLRs are then denoted by $\alpha_{ji}^t(k) := \alpha_{ji}(\pi(k))$, where $k=1,\ldots,d_m$, and represented by the ordered set

$$\alpha_{ji}^t = \left\{\alpha_{ji}^t(1),\ldots,\alpha_{ji}^t(d_m)\right\}, \alpha_{ji}^t(1) > ... > \alpha_{ji}^t(d_m). \quad (10)$$

On the other hand, the α messages associated with simplified edges will apply the ES strategy and only retain the largest symbol LLR (denoted as α_{ji}^m). Equipped with both the ST and the ES strategies, the BsP detector can build up a configuration set of possible TX symbols defined as follows:

$$\mathcal{B}(d_m,d_f) = \left\{\mathbf{s}_{\mathbf{k}/j} = \left[\mathbf{s}_{\mathbf{k}/j,d_f}^T;\ \tilde{\mathbf{s}}_{\mathbf{k}/j,d_f}^T\right]^T\right\}, \quad (11)$$

where $\mathbf{s}_{\mathbf{k}/j}$ represents the TX symbol vector excluding s_j, $\mathbf{s}_{\mathbf{k}/j,d_f}^T$ and $\tilde{\mathbf{s}}_{\mathbf{k}/j,d_f}^T$ denote the possible TX symbols corresponding to the chosen edges and simplified edges, respectively. Based on the configuration set $\mathcal{B}(d_m,d_f)$, the soft message β_{ij} is computed as Eq. (12). Then, the soft message α is given by

$$\alpha_{ji}(k) = \sum_{t=1,t\neq i}^{2N_r}\beta_{tj}(k). \quad (13)$$

When reaching the maximum detector iterations $Q_{\det}$, the BsP detector computes the symbol LLR for every TX symbol as

$$\gamma_j(k) = \sum_{i=1}^{2N_r}\beta_{ij}(k). \quad (14)$$

Finally, the BsP detector derives bit LLRs as Eq. (6). It is noted that although utilizing a large configuration set (e.g., $\mathcal{B}(2,2)$) improves the performance of the BsP detector, it introduces a significant burden in computational complexity. Therefore, without special explanation, we adopt the BsP with $\mathcal{B}(1,1)$ as the benchmark in this paper. More details about the BsP detector can be found in [19].

C. Offset Min-Sum LDPC Decoder

A well-known LDPC decoding algorithm is the BP algorithm. The BP decoder is usually illustrated through a Tanner graph, which involves two types of nodes: variable nodes (VNs) and check nodes (CNs), denoted as v_j and C_i, respectively. To the best of the authors' knowledge, the offset min-sum (OMS) [33] LDPC decoder is a widely-used BP decoder, known for its good balance between computational complexity and decoding performance. Moreover, the OMS decoding and BsP detection are both BP-based algorithms, which improves the efficiency of information exchange at the interface between the detector and decoder. Therefore, the OMS LDPC decoder is utilized in our turbo receiver. Specifically, the decoder first receives the *a priori* LLR information $L_{j,b}^{\mathrm{A1}}$. To enhance the clarity in the description of the OMS decoding algorithm, we define the bit mapping as $L_k^{\mathrm{A1}} = L_{j,b}^{\mathrm{A1}}, k=(j-1)\times\frac{M}{2}+b$. Therefore, the bit LLR ($L_{j,b}^{\mathrm{A1}}$) can be expressed as the k-th element L_k^{A1} in the bit LLR vector. Then, the LLR information is utilized to initialize VNs. In the OMS decoding algorithm, the soft messages in CNs and VNs are updated as

$$\begin{cases} r_{ij} = \prod\limits_{t\in N(i)/j}\operatorname{sgn}(q_{ti})\max\left\{\min\limits_{t\in N(i)/j}|q_{ti}|-\epsilon,0\right\},\\ q_{ji} = L_j^{A1} + \sum\limits_{t\in N(j)/i} r_{tj}, \end{cases} \quad (15)$$

where r_{ij} denotes the LLR message sent from C_i to v_j, q_{ji} represents the LLR message passed in the opposite direction, and $N(i)/j$ is the set of VNs connected to C_i excluding v_j. Additionally, ϵ is the offset factor, which can be tuned to optimize decoding performance. During each decoding iteration, the decoder computes the extrinsic LLR information for each source bit, expressed as

$$L_j^{\mathrm{E2}} = L_j^{\mathrm{A1}} + \sum_{t\in N(j)} r_{tj}. \quad (16)$$

Subsequently, the estimated results of source bits are obtained by hard decision, expressed as

$$\hat{b}_k = \begin{cases} 0 & \text{if } L_k^{\mathrm{E2}} \geq 0,\\ 1 & \text{otherwise.}\end{cases} \quad (17)$$

If the estimated bits satisfy the early stopping criterion based on parity checks or the maximum number of receiver iterations is reached, the receiver outputs the final estimated bit sequence. Otherwise, the decoder remaps the bit LLR L_k^{E2} into $L_{j,b}^{\mathrm{E2}}$, which is fed back to the detector for the next round of iteration.

III. Fixed-constellation BsP Detection

The FCBsP detection algorithm together with its corresponding turbo receiver is proposed in this section. First, two unexplored techniques: *fixed configuration set* and *approximate multi-user interference* are proposed to facilitate the proposed

FCBsP detector, resulting in a better trade-off between error performance and computational complexity compared with the BsP detector. Second, an information compensation strategy is proposed to address the missing counter-hypotheses issue [31] of the FCBsP-based turbo receiver, enhancing its performance with negligible complexity consumption. Finally, a decoding-first mechanism is designed for the FCBsP-based turbo receiver, sufficiently lowering its processing latency. The proposed FCBsP detection algorithm in the turbo receiver is also summarized in this section.

A. FCBsP Detection Algorithm

Although the aforementioned BsP detector shows effectiveness compared with the original BP detector, it still suffers from performance and complexity bottlenecks, especially when applied in MIMO turbo receivers. To address this issue, we propose the FCBsP detection algorithm, which utilizes the fixed configuration set and the approximate multi-user interference to lower the computational complexity of message updates in the BsP detector.

1) Fixed configuration set: For an FCBsP-based MIMO turbo receiver, the fixed configuration set is built based on the symbol LLRs (initial $\boldsymbol{\gamma}$ messages) of each SN, which are updated with the *a priori* information provided by the decoder. The updated symbol LLRs are considered as a reliability metric for constellation symbols, where a larger LLR value indicates higher reliability. Following this, the fixed configuration set is constructed by incorporating the n_m most reliable constellation symbols for each SN. Therefore, define $\mathcal{E}(n_m)$ as the fixed configuration set, and it can be built by

$$\mathcal{E}(n_m) = \Big\{\mathcal{E}_k(n_m) = \left[\mu_{\varphi_1}, \mu_{\varphi_2}, \ldots, \mu_{\varphi_{n_m}}\right] : \forall \boldsymbol{\varphi},\ \gamma_k(\varphi_1) > \cdots > \gamma_k(\varphi_{n_m})\Big\}, \tag{18}$$

where $\mathcal{E}_k(n_m)$, $k \in \{1, 2, ..., 2N_t\}$, denotes the subset corresponding to the k-th SN, and $\left[\mu_{\varphi_1}, \mu_{\varphi_2}, ..., \mu_{\varphi_{n_m}}\right]$ denotes the n_m most reliable constellation symbols derived from the updated $\boldsymbol{\gamma}$ messages corresponding to the k-th SN. Here note that in uncoded MIMO systems, the fixed configuration set $\mathcal{E}(n_m)$ can be built from the $\boldsymbol{\gamma}$ messages initialized by the preprocessor.

2) Approximate multi-user interference: For the proposed FCBsP detector, the received signal of the i-th FN can be expanded as

$$\begin{aligned} y_i &= h_{ij}s_j + \underbrace{\sum_{k=1,k\neq j}^{2N_t} h_{ik}s_k}_{\text{multi-user interference}} + n_i, \\ &= h_{ij}s_j + z_{ij} + n_i, \end{aligned} \tag{19}$$

where z_{ij} denotes the multi-user interference of y_i in terms of the j-th TX symbol. Empirically, more precise multi-user interference in (19) will result in more reliable $\boldsymbol{\beta}$ messages updated by this FN, thus achieving better error performance for the FCBsP detector. However, the optimal multi-user interference often requires an exhaustive search of possible TX symbols. Although the fixed configuration set $\mathcal{E}(n_m)$ introduced above has sufficiently reduced the number of the surviving candidates for each TX symbol, the exhaustive search will still lead to unaffordable computational complexity. To overcome this dilemma, the approximate multi-user interference is utilized for the message updates. More specifically, we denote the approximate multi-user interference as ϕ_{ij} to replace the z_{ij} in (19), and ϕ_{ij} is expressed as

$$\begin{cases} \phi_{ij} = \displaystyle\sum_{k=1,k\neq j}^{2N_t} h_{ik} \cdot \bar{s}_k, \\ \bar{s}_k = \displaystyle\sum_{\mu_{\varphi_t} \in \mathcal{E}_k(n_m)} \mu_{\varphi_t} \cdot P(s_k = \mu_{\varphi_t}). \end{cases} \tag{20}$$

where $\bar{s}_k$ represents the symbol mean of the k-th TX symbol and $P(s_k = \mu_{\varphi_t})$ is the *a priori* probability for the k-th TX symbol with $s_k = \mu_{\varphi_t}$. Given the incoming message α, $P(s_k = \mu_{\varphi_t})$ is updated as

$$P(s_k = \mu_{\varphi_t}) \approx \exp\left(\alpha_{ki}(\varphi_t) - \max_{\mu_{\varphi_m} \in \mathcal{E}_k(n_m)} \alpha_{ki}(\varphi_m)\right). \tag{21}$$

The detailed derivation of Eq. (21) is provided in Appendix. A.

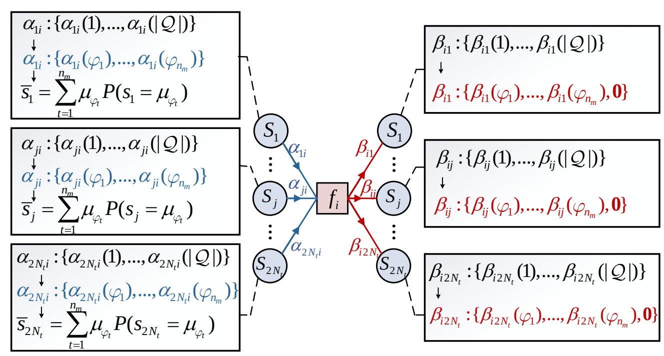


Fig. 3: Message updates of the FCBsP detector.

3) Message updates: Fig. 3 shows the message-updating process of the i-th FN for the proposed FCBsP detector. Before the message updates, a fixed configuration set $\mathcal{E}(n_m)$ is built based on the initial $\boldsymbol{\gamma}$ messages. Then, in each iteration, the incoming α_{ji} messages will be truncated, reserving only the n_m LLRs corresponding to the $\mathcal{E}_j(n_m)$. Based on these surviving symbol LLRs, the symbol mean $\bar{s}_j$ together with the approximate multi-user interference ϕ_{ij} is updated as Eq. (20). Therefore, the $\boldsymbol{\beta}$ messages of i-th FN are computed by

$$\beta_{ij}(\varphi_k) = \frac{(y_i - \phi_{ij} - h_{ij}\mu_1)^2}{2\sigma_n^2} - \frac{(y_i - \phi_{ij} - h_{ij}\mu_{\varphi_k})^2}{2\sigma_n^2}, \tag{22}$$

where $\mu_{\varphi_k} \in \mathcal{E}_j(n_m)$, $k \in \{1, ..., n_m\}$. It is noted that we utilize the variance of system noise σ_n^2 to replace the theoretical noise expressed as $\sum_{k=1,k\neq j}^{2N_t} h_{ik}^2 \cdot \{\sum_{t=1}^{n_m} \mu_{\varphi_t}^2 P(s_k = \mu_{\varphi_t}) - \bar{s}_k^2\} + \sigma_n^2$, which can reduce the complexity of the proposed FCBsP detector with negligible performance degradation. Compared with Eq. (12), the incoming message α_{ji} is omitted in Eq. (22) since it has been utilized to compute the symbol mean and the approximate multi-user interference. Here note that the message updates for β_{ij} are limited to the constellation symbols within $\mathcal{E}_j(n_m)$, and other un-updated β

messages are maintained as zero. As a result, β messages can be expressed as

$$\begin{aligned}\beta_{ij} = \{&[\beta_{ij}(\varphi_1), \dots, \beta_{ij}(\varphi_{n_m}), \mathbf{0}] : \\ &\mu_{\varphi_k} \in \mathcal{E}_j(n_m), k \in \{1, ..., n_m\}\}.\end{aligned} \tag{23}$$

Equipped with the fixed configuration set and the approximate multi-user interference, the proposed FCBsP has three main differences compared with the BsP detector. First, once the fixed configuration set is constructed, it will not be updated by the proposed FCBsP throughout the message-updating iterations. However, the BsP detector is required to update its configuration set at each iteration. Second, the proposed FCBsP detector merely updates n_m symbols within $\mathcal{E}_j(n_m)$ for each message β_{ij}, while the BsP detector has to compute all $|\mathcal{Q}|$ constellation symbols for their $\boldsymbol{\beta}$ messages. Finally, the proposed FCBsP detector computes the β messages by utilizing the approximate multi-user interference, instead of the exhaustive search for all possible TX symbols in the configuration set. Consequently, the proposed FCBsP detector can achieve a more favorable trade-off between complexity and performance compared to BsP. However, utilizing $\boldsymbol{\beta}$ messages as Eq. (23) to generate bit LLRs for the channel decoder yields the so-called missing counter-hypotheses issue [31, 34, 35], which will result in performance degradation for the FCBsP based MIMO turbo receiver. In the following subsection, we will introduce a new method to compensate for LLR information in our proposed FCBsP-based turbo receiver.

B. Information Compensation

In uncoded MIMO systems, the proposed FCBsP achieves a better complexity and performance trade-off than the BsP detector. However, in coded MIMO systems, directly using the $\boldsymbol{\gamma}$ messages from the proposed FCBsP detector to generate bit LLRs for the channel decoder may degrade performance due to missing the symbol LLRs outside the fixed configuration set (set as zero). This issue, known as missing counter-hypotheses [31], has been widely studied in tree-search detection algorithms [34, 35]. Inspired by literature [36], we propose an information compensation strategy to rebuild the missing symbol LLRs, thus mitigating the performance loss. After the proposed FCBsP detection reaches the maximum iterations, it will output the symbol LLRs as follows

$$\boldsymbol{\gamma}_j = \left[\gamma_j(\varphi_1), \gamma_j(\varphi_2), \cdots, \gamma_j(\varphi_{|\mathcal{Q}|})\right], \tag{24}$$

where $\gamma_j(\varphi_1) > \gamma_j(\varphi_2) > ... > \gamma_j(\varphi_{n_m})$, and $\gamma_j(\varphi_{n_m+1}) = \cdots = \gamma_j(\varphi_{|\mathcal{Q}|}) = 0$ denote the truncated messages which need to be compensated. For the reader's better understanding, we define $\left[\hat{\gamma}_j(\varphi_{n_m+1}), \hat{\gamma}_j(\varphi_{n_m+2}), \cdots, \hat{\gamma}_j(\varphi_{|\mathcal{Q}|})\right]$ as the reconstructed messages. Empirically, the probabilities of the truncated symbols corresponding to the $\hat{\gamma}_j(\varphi_{n_m+1}) \sim \hat{\gamma}_j(\varphi_{|\mathcal{Q}|})$ messages should decrease rapidly. Therefore, we utilize a geometric decreasing model to reconstruct the probabilities, which can not only match their decay feature but also enjoy a low complexity burden. Specifically, define θ as a descending factor, the probabilities corresponding to the truncated symbols can be computed by

$$P(s_j = \mu_{\varphi_k}) = \theta^{(k-n_m-1)} P(s_j = \mu_{\varphi_{n_m+1}}), \tag{25}$$

where $\theta \in [0, 1)$, and $k \in \{n_m + 2, \dots, |\mathcal{Q}|\}$. According to Eq. (25), we can easily derive the truncated messages as

$$\hat{\gamma}_j(\varphi_k) = \hat{\gamma}_j(\varphi_{n_m+1}) + (k - n_m - 1)\log\theta, \tag{26}$$

where $k \in \{n_m + 2, \dots, |\mathcal{Q}|\}$. Eq. (26) shows that the entire reconstructed messages can be computed from the $\hat{\gamma}_j(\varphi_{n_m+1})$. Based on the conditions above, $\hat{\gamma}_j(\varphi_{n_m+1})$ can be derived as follows.

$$\begin{aligned}
&\sum_{k=1}^{n_m} P(\mu_{\varphi_k}) + \sum_{k=n_m+1}^{|\mathcal{Q}|} P(\mu_{\varphi_k}) \overset{(a)}{=} 1, \\
&\frac{P(\mu_{\varphi_{n_m+1}})\left(1-\theta^{(|\mathcal{Q}|-n_m)}\right)}{1-\theta} \overset{(b)}{=} 1 - \sum_{k=1}^{n_m} P(\mu_1) e^{\gamma_j(\varphi_k)}, \\
&\log\frac{P(\mu_{\varphi_{n_m+1}})}{P(\mu_1)} \overset{(c)}{=} \log\left(\frac{1}{P(\mu_1)} - \sum_{k=1}^{n_m} e^{\gamma_j(\varphi_k)}\right) - \log b, \\
&\hat{\gamma}_j(\varphi_{n_m+1}) \overset{(d)}{=} \log\left(\sum_{k=n_m+1}^{|\mathcal{Q}|} e^{\gamma_j(\varphi_k)}\right) - \log b, \\
&\hat{\gamma}_j(\varphi_{n_m+1}) \overset{(e)}{\approx} \gamma_j(\varphi_{n_m+1}) - \log b,
\end{aligned} \tag{27}$$

in which we replace $P(s_j = \mu_{\varphi_k})$ with $P(\mu_{\varphi_k})$ for simplicity, (b) sums the last $|\mathcal{Q}| - n_m$ symbol values, (c) is a log-form of (b) with $b = \frac{\left(1-\theta^{(|\mathcal{Q}|-n_m)}\right)}{1-\theta}$, (d) is obtained due to $\frac{1}{P(\mu_1)} = \sum_{k=1}^{|\mathcal{Q}|} e^{\gamma_j(\varphi_k)}$, and (e) is derived through the max-log approximation. It is noted that although Eq. (27) provides a theoretical derivation for the $\hat{\gamma}_j(\varphi_{n_m+1})$, it requires to reserving an additional $\gamma_j(\varphi_{n_m+1})$ messages for the FC-BsP detector, which will result in a higher complexity. However, we can regard the $\gamma_j(\varphi_{n_m})$ as the first truncated messages, and compute $\hat{\gamma}_j(\varphi_{n_m})$ according to Eq. (27). Then, the entire truncated γ messages can be reconstructed with Eq. (26).

C. Decoding-First Turbo Receiver

To sufficiently reduce the process latency of the FCBsP-based MIMO turbo receiver, a decoding-first mechanism is proposed, which exploits the initial results from the preprocessor more effectively. Fig. 4 shows the system diagram of the decoding-first FCBsP-based MIMO turbo receiver, in which the initial results of TX symbols are directly converted into bit LLRs, computed as

$$\begin{aligned}
L_{j,b}^{\text{init1}} &= \log\left(\frac{\sum_{\mu_k\in\mathcal{Q}_b^1}\exp(-\frac{\|\mu_k-\hat{\mathrm{s}}_{\text{init},j}\|^2}{2\sigma_{\text{init},j}^2})}{\sum_{\mu_k\in\mathcal{Q}_b^0}\exp(-\frac{\|\mu_k-\hat{\mathrm{s}}_{\text{init},j}\|^2}{2\sigma_{\text{init},j}^2})}\right) \\
&\overset{\text{LSE}}{\approx} \frac{\min\limits_{\mu_k\in\mathcal{Q}_b^0}\|\mu_k-\hat{\mathrm{s}}_{\text{init},j}\|^2 - \min\limits_{\mu_k\in\mathcal{Q}_b^1}\|\mu_k-\hat{\mathrm{s}}_{\text{init},j}\|^2}{2\sigma_{\text{init},j}^2},
\end{aligned} \tag{28}$$

where $L_{j,b}^{\text{init1}}$ denotes the initial bit LLR for the b-th bit in $\hat{\mathrm{s}}_{\text{init},j}$. After de-interleaving, the initial bit LLR ($L_{j,b}^{\text{init2}}$) is directly sent to the LDPC decoder as *a priori* information, bypassing the FCBsP detector. If the results of the decoder are correct, the decoding-first turbo receiver will output the estimated bit vector $\hat{\mathrm{b}}$. Otherwise, the FCBsP detector will be

This article has been accepted for publication in IEEE Transactions on Wireless Communications. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TWC.2025.3636908

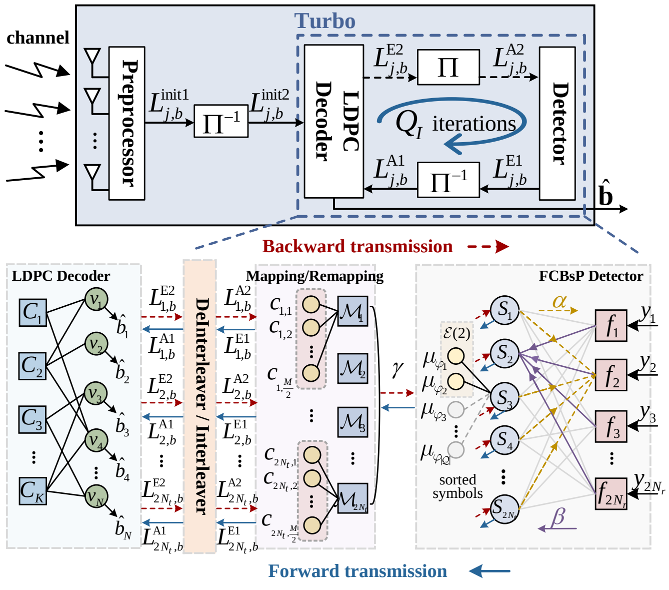


Fig. 4: System diagram of the proposed decoding-first FCBsP-based MIMO turbo receiver.

Algorithm 1: FCBsP Detection Algorithm

Input : $\mathbf{y}$, $\mathbf{H}$, $\mathcal{Q}$, Q_{det}, $\mathbf{L}^{\text{A2}}$, n_m, θ
Output : bit LLRs $\mathbf{L}^{\text{E1}}$

for $j \leftarrow 1$ **to** $2N_t$ **do** // initialize $\boldsymbol{\gamma}$ and $\boldsymbol{\alpha}$
  for $k \leftarrow 1$ **to** $|\mathcal{Q}|$ **do**
    initialize $\gamma_j(k)$ with (7);
    for $i \leftarrow 1$ **to** $2N_r$ **do**
      $\alpha_{ji}(k) = \gamma_j(k)$;
build up $\mathcal{E}(n_m)$ as (18);
for Iter $\leftarrow 1$ **to** Q_{det} **do**
  for $k \leftarrow 1$ **to** n_m **do**
    $\forall \mu_{\varphi_k} \in \mathcal{E}_j(n_m)$:
    for $i \leftarrow 1$ **to** $2N_r$ **do** // update $\boldsymbol{\beta}$
      for $j \leftarrow 1$ **to** $2N_t$ **do**
        update $P(s_j)$ with (21);
        compute ϕ_{ij} with (20);
        update $\boldsymbol{\beta}_{ij}$ as (22) ;
    for $j \leftarrow 1$ **to** $2N_t$ **do** // update $\boldsymbol{\gamma}$
      update $\gamma_j(\varphi_k)$ with (14);
      for $i \leftarrow 1$ **to** $2N_r$ **do** // update $\boldsymbol{\alpha}$
        update $\alpha_{ji}(\varphi_k)$ with (13);
for $k \leftarrow n_m + 1$ **to** $|\mathcal{Q}|$ **do** // info compensate
  for $j \leftarrow 1$ **to** $2N_t$ **do**
    compute $\gamma_j(\varphi_k)$ as (27) and (26);
compute $\mathbf{L}^{\text{E1}}$ with (6).

activated by the *a priori* information provided by the decoder to build the fixed configuration set. Compared to traditional MIMO turbo receivers (Fig. 1), the proposed decoding-first turbo receiver can achieve significant complexity reduction, especially in high SNR scenarios. Fig. 4 also illustrates the detailed message flow in the turbo receiver, which includes "backward" and "forward" transmission. The messages are propagated through the following four major modules: LDPC decoder, interleaver/deinterleaver, mapping/remapping, and FCBsP detector. These modules collectively form a joint factor graph for the iterative message updates in the turbo receiver.

Based on the proposed methods, our proposed FCBsP detection is summarized in Alg. 1. Specifically, after the FCBsP detector receives the *a priori* information, $\boldsymbol{\gamma}$ and $\boldsymbol{\alpha}$ messages are initialized. Then, the fixed configuration set $\mathcal{E}(n_m)$ is derived. During each iteration, $\boldsymbol{\beta}$ messages are updated based on the symbol probability $P(s_j)$ and the approximate multi-user interference ϕ_{ij} with $\mathcal{E}(n_m)$. Then, $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ messages are computed. When the maximum number of iterations Q_{det} is reached, the proposed FCBsP detection stops message updates and provides final symbol LLRs $\boldsymbol{\gamma}$. Finally, after compensating for the information of truncated symbols, the bit LLRs are computed and output.

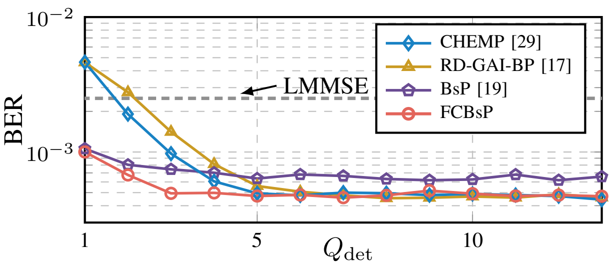
IV. Convergence and Complexity Analysis

This section presents convergence and complexity analyses to evaluate the performance of our proposed FCBsP detector in uncoded and turbo MIMO systems.

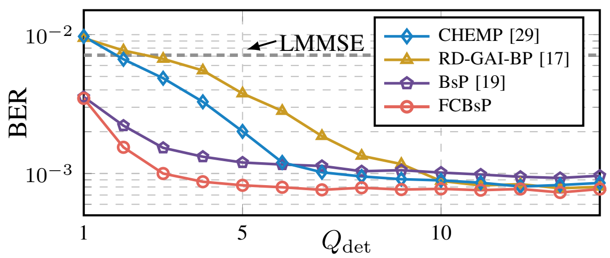
A. Convergence Analysis

The convergence of the FCBsP detector is analyzed in both uncoded and turbo MIMO systems. Note that the parameter settings are consistent with those for error performance, as specified in Section V-A. In the uncoded MIMO system, we compare the convergence speed of the FCBsP detector with the channel hardening-exploiting message passing (CHEMP) [29], RD-GAI-BP [17], and BsP [19] detectors. In Fig. 5, we investigate the convergence behavior of the proposed FCBsP in the following two MIMO scenarios: 32×12 with 64-QAM and 128×64 with 256-QAM. The results demonstrate that the proposed FCBsP enjoys faster convergence speed compared to its counterparts across various MIMO scenarios. Specifically, in 32×12 MIMO scenario as demonstrated in Fig. 5(a), the FCBsP detector converges within $Q_{\text{det}} = 3$ iterations, while both CHEMP and the RD-GAI-BP require $Q_{\text{det}} = 6$ iterations to converge. In contrast, in a large-scale 128×64 MIMO system (Fig. 5(b)), all detectors demand more iterations to converge. In this scenario, the BsP and the proposed FCBsP both converge after $Q_{\text{det}} = 5$ iterations, whereas CHEMP and RD-GAI-BP both require $Q_{\text{det}} = 10$ iterations to reach their converged points. Based on these observations, the detector iterations for the CHEMP and the RD-GAI-BP are both set as $Q_{\text{det}} = 10$, while the BsP and our proposed FCBsP are set as $Q_{\text{det}} = 5$ in subsequent uncoded MIMO simulations to ensure their convergence in various MIMO scenarios.

The convergence performance of the proposed FCBsP IDD receiver is compared with the BsP and CHEMP IDD receivers in a 5G $(1440, 1152)$ LDPC coded MIMO system, with an offset factor of $\epsilon = 0.25$ for the OMS decoding algorithm.



(a) 32×12, 64-QAM, @SNR=16



(b) 128×64, 256-QAM, @SNR=20

Fig. 5: Convergence performance of the proposed FCBsP detector in (a) 32×12, 64-QAM and (b) 128×64, 256-QAM uncoded MIMO system under Rayleigh channels.

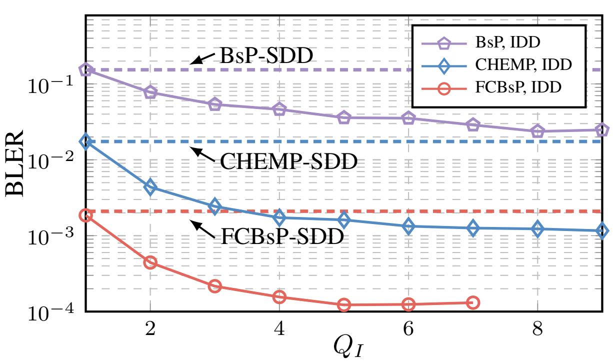
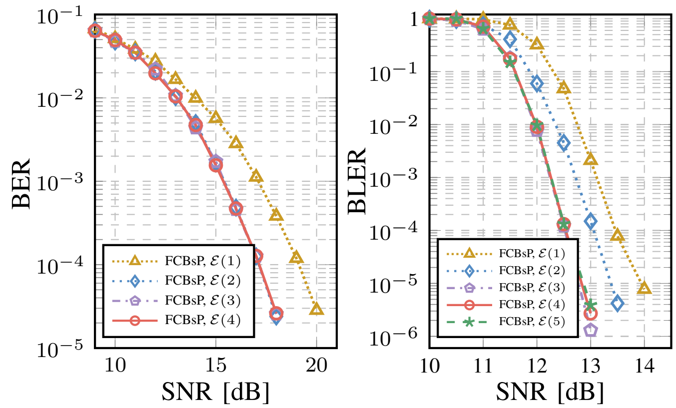


Fig. 6: Convergence performance of the proposed FCBsP IDD receiver in the 5G $(1440, 1152)$ LDPC coded 32×12 64-QAM MIMO system under Rayleigh channels at SNR $= 12.5$ dB.



(a) uncoded MIMO systems (b) turbo MIMO systems

Fig. 7: Fixed configuration set analysis in 32×12, 64-QAM uncoded and turbo MIMO systems under Rayleigh channels.

Fig. 6 presents the block error rate (BLER) performance with respect to the number of turbo iterations (Q_I) in a 32×12, 64-QAM MIMO system. The figure also involves the corresponding SDD receivers as a reference. The results indicate that our proposed FCBsP IDD receiver enjoys the fastest convergence speed among the compared IDD receivers, requiring only $Q_I = 5$ for convergence. However, the BsP and the CHEMP IDD receivers converge at about $Q_I = 8$. Considering the relatively high latency of turbo receivers, Q_I is set as 5 for all IDD receivers for subsequent simulations. From the convergence results, it can be observed that our proposed FCBsP detection offers superior convergence performance in both uncoded and turbo MIMO scenarios.

Fig. 7 illustrates the error performance of the FCBsP detector equipped with different fixed configuration sets in (a) uncoded and (b) turbo 32×12 MIMO systems with 64-QAM modulation. The results reveal that the FCBsP detector achieves better error performance with larger fixed configuration sets in both uncoded and turbo MIMO scenarios. However, the FCBsP detector in turbo MIMO systems requires a larger fixed configuration set to achieve convergence. Specifically, the proposed FCBsP detector configured with $\mathcal{E}(2)$ achieves near-optimal performance in the uncoded MIMO system, which can significantly save computational complexity, particularly in high-order modulation MIMO scenarios. Note that the error performance of the proposed FCBsP detector with $\mathcal{E}(1)$ cannot be further improved by increasing the maximum number of iterations. With only a single constellation symbol in the fixed configuration set, the symbol mean cannot be effectively updated during iterations, which significantly limits the performance gains from additional iterations. Furthermore, setting the maximum number of iterations to $Q_{\text{det}} = 5$ ensures that the proposed FCBsP detector with different $\mathcal{E}(n_m)$ configurations reaches convergence. In turbo MIMO systems, the FCBsP IDD receiver needs a larger fixed configuration set as $\mathcal{E}(3)$ to ensure convergence. This is because a smaller fixed configuration set may exacerbate error propagation in turbo MIMO systems. Therefore, turbo MIMO systems require the proposed FCBsP detector to incorporate a larger fixed configuration set to ensure robust and reliable performance.

B. Complexity Analysis

In this subsection, the computational complexity of our proposed FCBsP detector is evaluated by analyzing the involved real operations (multiplications, additions, exponentiations, and comparisons). Since all turbo receivers employ the same channel coding and decoding schemes, we focus on complexity comparisons of MIMO detectors. Table. I summarizes the computational complexity of our proposed FCBsP detectors and other detectors. The $\mathcal{O}$ notation expresses the quantity of each operation in the table. We begin by analyzing the complexity of the BsP. The table demonstrates the complexity of the BsP equipped with $\mathcal{B}(1,1)$. The dominant complexity of the BsP arises from the computation of $\boldsymbol{\beta}$ messages, which approximately involves $\mathcal{O}\left(8Q_{\text{det}}N_rN_t|\mathcal{Q}|\right)$ multiplications and

TABLE I
COMPLEXITY COMPARISONS OF DIFFERENT DETECTORS.

Detectors	Multiplications	Additions	Exponentiations	Comparisons
RD-GAI-BP [17]	$\mathcal{O}(12Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)+\chi^{\dagger}$	$\mathcal{O}(16Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)+\chi$	$\mathcal{O}(4Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)$	0
BsP [19]	$\mathcal{O}(8Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)+\chi$	$\mathcal{O}(8Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)+\chi$	0	$\mathcal{O}(4Q_{\text{det}}N_rN_t\lvert\mathcal{Q}\rvert)$
CHEMP [29]	$\mathcal{O}(6Q_{\text{det}}N_t\lvert\mathcal{Q}\rvert)+\chi$	$\mathcal{O}(8Q_{\text{det}}N_t\lvert\mathcal{Q}\rvert)+\chi$	$\mathcal{O}(2Q_{\text{det}}N_t\lvert\mathcal{Q}\rvert)$	0
EP [8]	$\mathcal{O}\left(Q_{\text{det}}(\frac{16}{3}N_t^3+8N_t\lvert\mathcal{Q}\rvert)\right)$	$\mathcal{O}\left(Q_{\text{det}}(\frac{16}{3}N_t^3+10N_t\lvert\mathcal{Q}\rvert)\right)$	$\mathcal{O}(2Q_{\text{det}}N_t\lvert\mathcal{Q}\rvert)$	0
FCBsP	$\mathcal{O}(12Q_{\text{det}}N_rN_tn_m)+\chi$	$\mathcal{O}(16Q_{\text{det}}N_rN_tn_m)+\chi$	$\mathcal{O}(4Q_{\text{det}}N_rN_tn_m)$	$\mathcal{O}(2N_t\lvert\mathcal{Q}\rvert)$

† : $\chi=\mathcal{O}(16/3N_t^3)$ denotes the complexity contributed by LMMSE initialization.

additions. Additionally, the BsP also requires $\mathcal{O}\left(16/3N_t^3\right)$ multiplications and additions for LMMSE initialization. The BsP also involves $\mathcal{O}(4Q_{\text{det}}N_rN_t|\mathcal{Q}|)$ comparisons for sorting $\boldsymbol{\alpha}$ messages in each iteration. As a low-complexity version of the BsP, the proposed FCBsP only updates $\boldsymbol{\beta}$ messages for a fixed set of constellation symbols, thereby reducing the number of updated constellation symbols from $|\mathcal{Q}|$ to n_m. In the process of the approximating multi-user interference, the proposed FCBsP introduces extra $\mathcal{O}(4Q_{\text{det}}N_rN_tn_m)$ multiplications and additions to compute ϕ_{ij}, as well as $\mathcal{O}(4Q_{\text{det}}N_rN_tn_m)$ additions and exponentiations to compute the probability $P(s_k=\mu_{\varphi_t})$. Consequently, the FCBsP requires a total of $\mathcal{O}(12Q_{\text{det}}N_rN_tn_m)+\mathcal{O}\left(16/3N_t^3\right)$ multiplications, $\mathcal{O}(16Q_{\text{det}}N_rN_tn_m)+\mathcal{O}\left(16/3N_t^3\right)$ additions, and $\mathcal{O}(4Q_{\text{det}}N_rN_tn_m)$ exponentiations. Moreover, since the proposed FCBsP only needs to sort $\boldsymbol{\gamma}$ messages once before starting the iterations, the complexity of comparison operations is $\mathcal{O}(2N_t|\mathcal{Q}|)$. In contrast, the RD-GAI-BP does not involve any comparison operation but requires updating all $|\mathcal{Q}|$ constellation symbols, which significantly increases its computational complexity. The CHEMP offers relatively low computational complexity, particularly in high-modulation scenarios, due to its proposed channel hardening strategy. Table. I also involves the computational complexity of the EP detector [8]. Specifically, the EP detector requires $\frac{16}{3}N_t^3+8N_t|\mathcal{Q}|$ multiplications, $\frac{16}{3}N_t^3+10N_t|\mathcal{Q}|$ additions, and $2N_t|\mathcal{Q}|$ exponentiations for each iteration. These results indicate that the EP detector incurs relatively high computational complexity when faced with high-dimensional MIMO systems.

To ensure a fair comparison of computational complexity, the FxPOs [20] is introduced as a unified metric to measure the complexity of each detector. The FxPOs are computed as

$$\text{FxPOs}=k^2\text{Mul.}+k\text{Add.}+mk^2\text{Exp.},\tag{29}$$

where k and m denote the bit widths for addition and exponentiation operations, respectively. Eq. (29) expresses the bit width required for the quantization of each real operation. Note that the comparison operation requires the same bit width as the addition operation, as it can be implemented through a subtraction operation. Therefore, the FxPOs provides a common basis to compare the complexity of different detectors. Fig. 8 demonstrates the FxPOs results for medium- (8×4) and large-scale (128×64) uncoded MIMO systems with 256-QAM modulation. The bit width is set as $k=14$ and $m=8$ following [20]. The number of iterations is set in accordance with the convergence results in Section IV-A. The proposed FCBsP is equipped with $\mathcal{E}(2)$ for complexity comparison. From Fig. 8, several key observations can be drawn. First, our proposed FCBsP shows a significant computational advantage over the RD-GAI-BP, with more than 90% reduction in FxPOs in both medium- and large-scale MIMO scenarios. Second, the proposed FCBsP achieves 58% and 47% complexity reduction compared to the BsP in the 8×4 and 128×64 uncoded MIMO systems, respectively. In addition, although the proposed FCBsP exhibits relatively higher complexity in the 128×64 massive MIMO system compared to the CHEMP, it provides 35% complexity reduction in the 8×4 MIMO system. Finally, compared with the EP detector, the proposed FCBsP shows comparable complexity in the 8×4 MIMO system, but achieves 48% complexity reduction when applied to the 128×64 MIMO scenario. Moreover, the EP detector requires computing matrix inversion during each iteration, which sufficiently limits its hardware efficiency in massive MIMO systems.

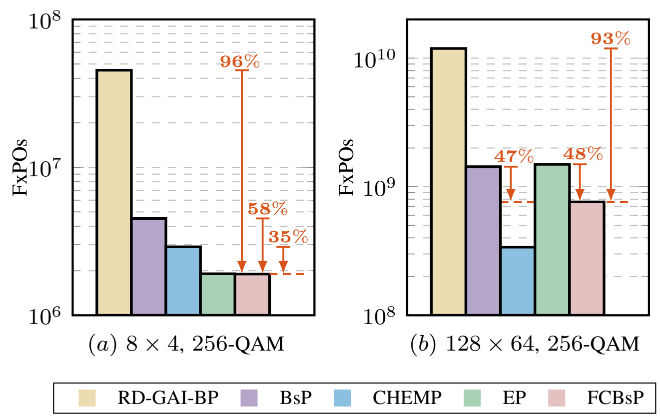


Fig. 8: Computational complexity comparisons of different detectors in (a) 8×4 and (b) 128×64 uncoded MIMO systems with 256-QAM modulation.

V. NUMERICAL RESULTS

In this section, numerical results are provided to demonstrate the performance of the proposed FCBsP detection in

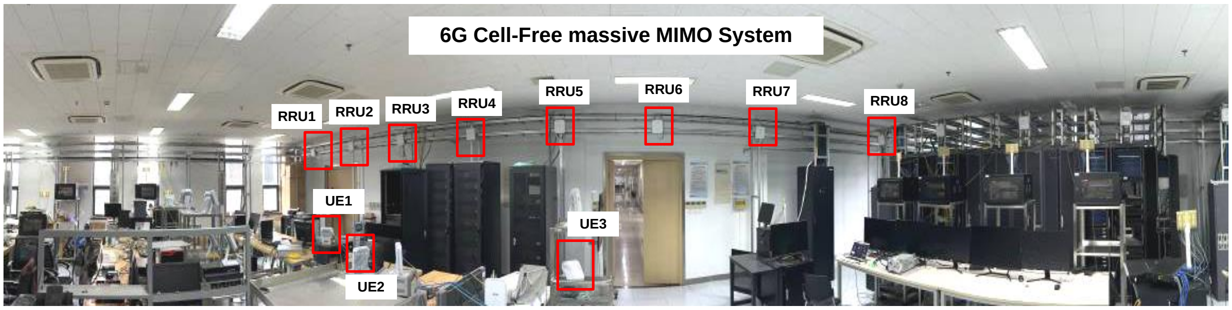


Fig. 9: The 6G Cell-Free massive-MIMO system for acquiring practical channels.

various MIMO scenarios. First, the parameter settings for both uncoded and LDPC-coded MIMO scenarios are detailed. Second, the error performance of the proposed FCBsP detection is analyzed under Rayleigh channels. Finally, the error performance of our proposed scheme under practical channels is evaluated.

A. Parameter Setting

The SNR of MIMO systems is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\tilde{E}_s N_t N_r}{\sigma_n^2 N_t R_c M}\right), \tag{30}$$

where $\tilde{E}_s$ is the average symbol energy, which is set to 1 in this paper. To thoroughly evaluate the performance of our proposed FCBsP detector and its corresponding receiver, we use different performance metrics for uncoded and coded MIMO systems. Specifically, BER is utilized as the performance metric for uncoded MIMO systems, while BLER is used for coded MIMO systems, where each block consists of K uncoded bits. Both Rayleigh and practical channels are applied to simulate the performance of MIMO systems. Under the Rayleigh channel model, each element of the channel matrix $\tilde{\mathbf{H}}$ follows the Gaussian distribution with zero mean and unit variance, i.e., $\tilde{h}_{ij} \sim \mathcal{CN}(0,1), \tilde{h}_{ij} \in \tilde{\mathbf{H}}$. For the practical channel, the channel coefficients are captured from our realistic 6G scalable Cell-Free massive MIMO (CF-mMIMO) system [37], as illustrated in Fig. 9. Specifically, we consider an uplink scenario with three four-layer user equipment (UEs) and a gNodeB equipped with eight distributed Remote Radio Units (RRUs), where each RRU is equipped with four receive antennas. Consequently, the practical channel matrix has the dimension of 32×12.

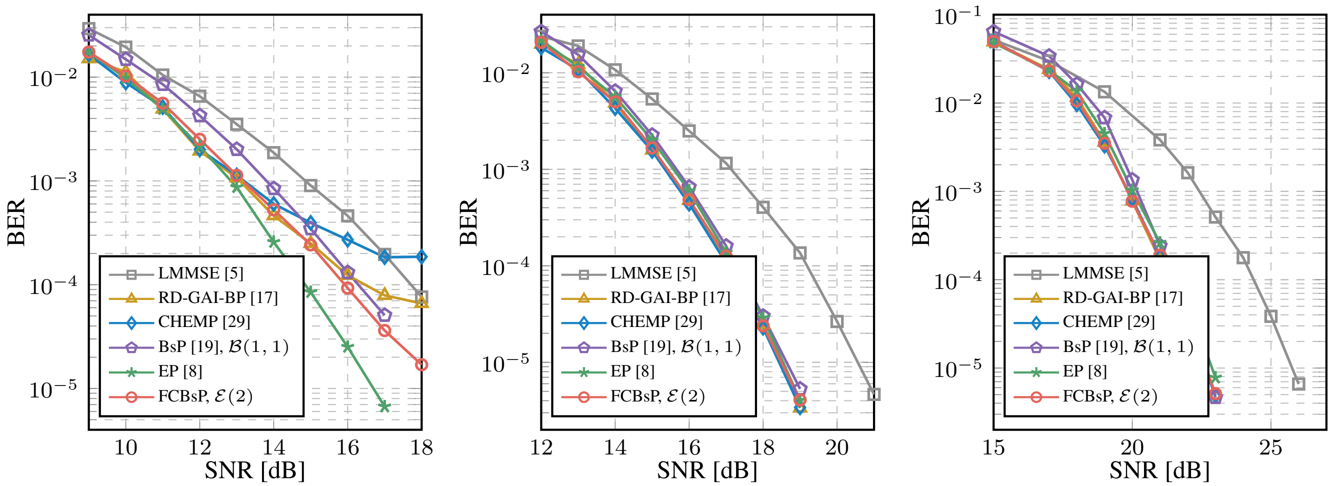
For uncoded MIMO scenarios, the error performance of our proposed FCBsP detector is compared with the RD-GAI-BP, CHEMP, and BsP detectors. The EP detector is also involved in comparison as a near-optimal benchmark. The RD-GAI-BP, CHEMP, BsP, and proposed FCBsP are all initialized with the LMMSE. To ensure convergence for each detector, the number of iterations is set as $Q_{\mathrm{det}} = 10$ for the RD-GAI-BP and CHEMP, and $Q_{\mathrm{det}} = 5$ for the BsP, EP, and proposed FCBsP. The damping factor of the CHEMP and RD-GAI-BP in all simulation scenarios are both set to $d = 0.33$, following references [29] and [17], respectively. The BsP is equipped with $\mathcal{B}(1,1)$. The proposed FCBsP is configured with $\mathcal{E}(2)$ and the descending factor is set to $\theta = 0.8$.

In coded MIMO scenarios, the RD-GAI-BP SDD and IDD receivers are excluded from performance comparisons due to their prohibitive complexity and latency. The proposed FCBsP SDD/IDD receivers are compared with the BsP and CHEMP SDD/IDD receivers. The performance of the LMMSE SDD receiver is also involved as a benchmark due to its wide utilization in practical receivers. Our proposed FCBsP detection is equipped with $\mathcal{E}(4)$ and the descending factor is also set to $\theta = 0.8$. For all receivers, the detection and decoding iteration are set as $Q_{\mathrm{det}} = 3$ and $Q_{\mathrm{dec}} = 25$, respectively. The turbo iteration of all IDD receivers is set to $Q_I = 5$ following the convergence results. The offset factor in the OMS decoding algorithm is set as $\epsilon = 0.25$. The Modulation and Coding Schemes (MCS) for coded MIMO scenarios adhere to the 5G NR standard. Specifically, MCS indices $I_{MCS} = 25$ (64-QAM, $R_c \approx 0.8$) and $I_{MCS} = 19$ (64-QAM, $R_c \approx 0.5$) from MCS Table 1 in [38] are adopted in simulations.

B. Performance Evaluation under Rayleigh Channels

This subsection evaluates the error performance of the proposed FCBsP detector in both uncoded and LDPC-coded MIMO systems under Rayleigh channels. Additionally, the performance advantages of our proposed decoding-first mechanism and information compensation strategy are also analyzed.

1) Uncoded MIMO Scenarios: Fig. 10 illustrates the BER performance of the proposed FCBsP detector in various MIMO scenarios under Rayleigh channels. Specifically, Fig. 10(a) compares the error performance in the medium-scale 8×4, 16-QAM MIMO scenario. Both the RD-GAI-BP and CHEMP exhibit significant error floors at high SNRs. The proposed FCBsP detector achieves an SNR gain of about 1.5 dB and 0.7 dB over the LMMSE and the BsP at BER $= 10^{-3}$, respectively. Fig. 10(b) and Fig. 10(c) present the BER performance comparisons for larger MIMO configurations, specifically 32×12 with 64-QAM and 128×64 with 256-QAM, respectively. It is observed that all aforementioned message-passing detectors offer near-optimal error performance in high-SNR regions. However, the BsP suffers about 0.5 dB performance loss at the BER of 10^{-2} compared to the proposed FCBsP at relatively low SNR regions, due to its truncated symbol search space for message updating. Moreover, our proposed FCBsP detector



(*a*) 8×4, 16-QAM (*b*) 32×12, 64-QAM (*c*) 128×64, 256-QAM

Fig. 10: BER performance of the proposed FCBsP detector in (a) 8×4, 16-QAM, (b) 32×12, 64-QAM, and (c) 128×64, 256-QAM uncoded MIMO scenarios under Rayleigh channels.

achieves over 2.0 dB SNR gain compared to the LMMSE detector. In addition, compared with the EP detector, the proposed FCBsP detector exhibits a slight performance loss in the 8×4 MIMO system. In contrast, for the 32×12 and 128×64 MIMO scenarios, both detectors converge to near-optimal performance. However, it should be noted that the FCBsP detector offers significant complexity advantages in massive MIMO systems.

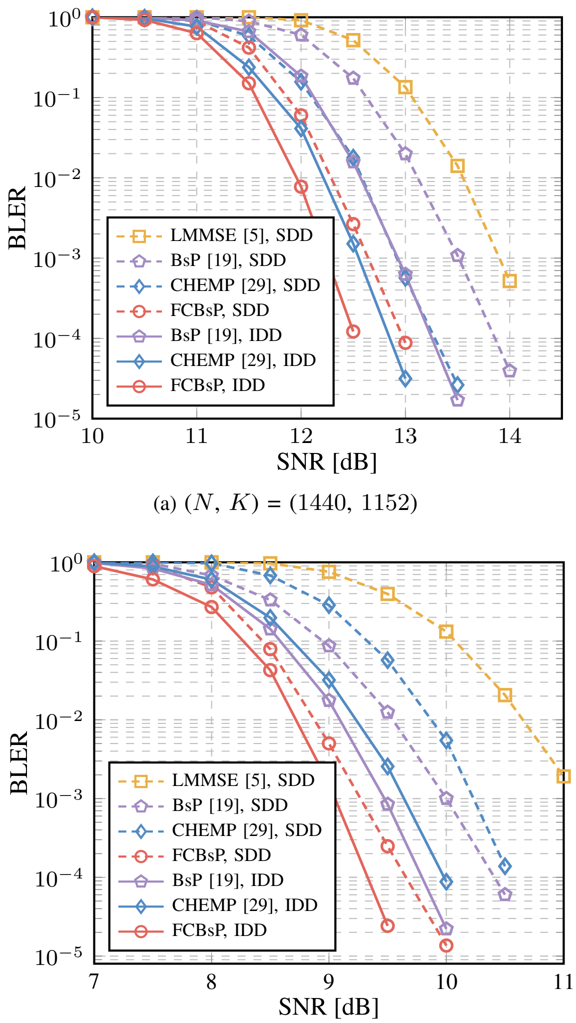
2) Coded MIMO Scenarios: Fig. 11 illustrates the error performance of our proposed FCBsP receivers across 32×12 64-QAM LDPC coded MIMO systems over Rayleigh channels. The performance evaluation considers two different code rate scenarios: a high-rate $(1440, 1152)$ code for Fig. 11(a) and a medium-rate $(1440, 720)$ code for Fig. 11(b). In the high-rate MIMO scenario, the receiver gains from the decoding are limited. In this case, receivers equipped with relatively high-performance detectors exhibit significant advantages. Fig. 11(a) reveals that our proposed FCBsP SDD receiver earns 0.9 and 0.3 dB SNR gains compared to the BsP and CHEMP SDD receivers at BLER $= 10^{-3}$, respectively. This indicates the superior error performance of the FCBsP detector in coded MIMO systems. Our proposed FCBsP IDD receiver provides a 0.4 dB SNR gain over its SDD counterpart and a 0.7 dB gain over the BsP IDD receiver, demonstrating its excellent efficiency and reliability in information exchange. In the medium-rate MIMO scenario shown in Fig. 11(b), decoding gains mitigate performance gaps across receivers. However, our proposed FCBsP IDD receiver still earns 0.4 and 0.6 dB advantages over the BsP and CHEMP IDD receivers, respectively. To ensure a fair performance comparison, the maximum number of iterations for detectors and turbo receivers is set to $Q_{\text{det}} = 3$ and $Q_I = 5$, respectively. It is observed that the performance gap between our proposed FCBsP turbo receiver and other turbo receivers is larger than that exhibited in the uncoded system, where detectors are equipped with different maximum iteration numbers to reach their performance convergence. From the error rate results, it can be concluded that the proposed FCBsP detector demonstrates effective error performance in both uncoded and coded MIMO systems under Rayleigh channels.

TABLE II
STATISTICAL ANALYSIS OF THE FCBSP RECEIVER.

Receiver Outcome	Decoding Succeeds		Decoding Fails
	case-1*	*case*-2⋆	*case*-3†
Sample Blocks	$10,000$		
SNR [dB]	12		
Blocks Number	**6420**	3475	105
Proportion‡	**64.88%**	35.12%	—

* : *case*-1 – the receiver successfully decodes bypassing the FCBsP detector.
⋆ : *case*-2 – the receiver successfully decodes after turbo processing.
† : *case*-3 – the receiver fails to recover the original bits.
‡ : The proportion is computed on successful-decoding sample blocks.

In Section III-C, we claim that the proposed decoding-first IDD receiver achieves reduced computational complexity compared to the traditional detection-first IDD receiver. To support this claim, we provide a statistical case study. In detail, we generate M_s source codes as sample blocks. At the receiver, each sample derives one of the following three outcomes: *case*-1 indicates that the receiver successfully decodes bypassing the FCBsP detector; *case*-2 represents the receiver successfully decodes after iterative turbo processing; *case*-3 denotes that the receiver fails to recover the original bits. Table. II demonstrates the statistical results of our proposed FCBsP IDD receiver in a 32×12 64-QAM LDPC-coded MIMO system with $M_s = 10000$ at 12 dB SNR. The table reveals that *case*-1 accounts for about 64.88% of all successful decoding samples, which shows that a substantial proportion of samples in the proposed decoding-first IDD receiver successfully bypassed the turbo processing, significantly reducing the computational complexity.



(a) $(N,\ K)$ = (1440, 1152)

(b) $(N,\ K)$ = (1440, 720)

Fig. 11: BLER performance of the proposed FCBsP-based receiver in 32×12 64-QAM LDPC-coded MIMO systems under Rayleigh channels.

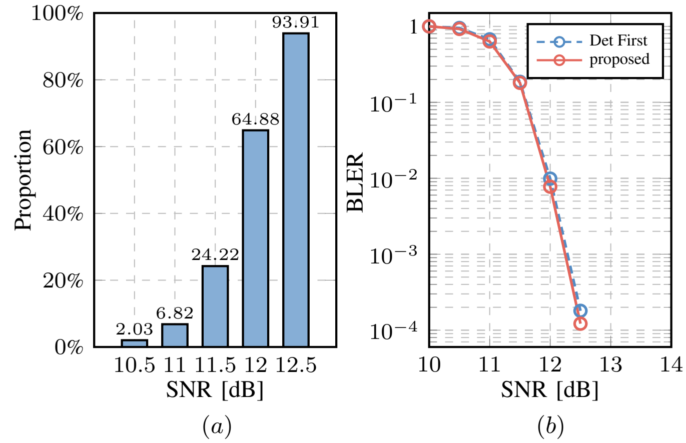
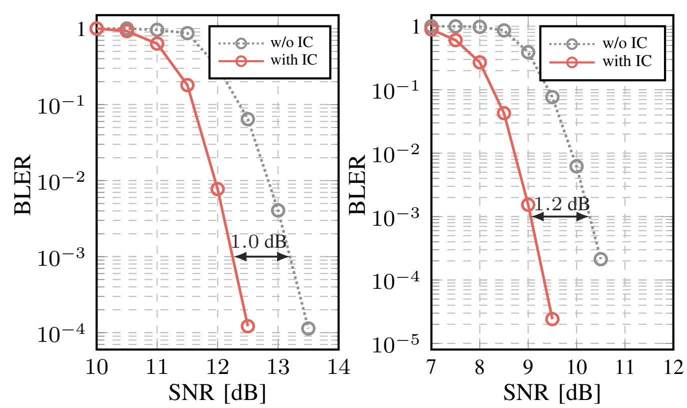


Fig. 12: (a) Proportion of *case*-1 in successful decoding samples under various SNRs and (b) BLER performance comparisons between the detection-first and decoding-first mechanism for the proposed FCBsP-based receivers in the 5G (1440, 1152) LDPC coded 32×12 64-QAM MIMO system.



(a) $(N,\ K)$ = (1440, 1152) (b) $(N,\ K)$ = (1440, 720)

Fig. 13: BLER performance comparisons of the proposed information compensation strategy for the FCBsP-based IDD receiver in 32×12 64-QAM LDPC-coded MIMO systems.

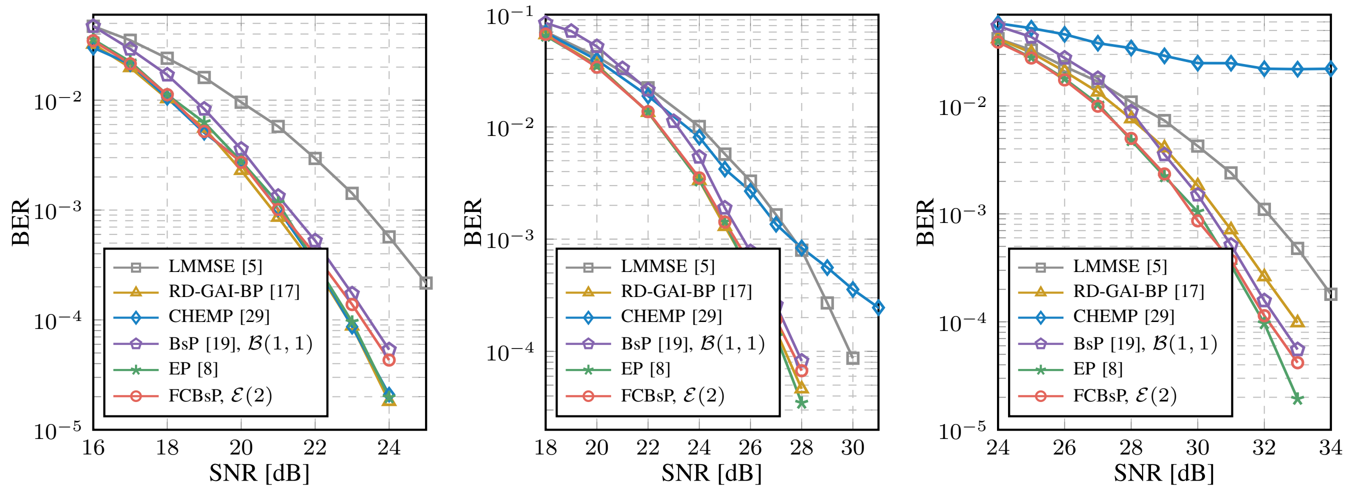
Fig. 12(a) presents the proportion of *case*-1 in all successful-decoding samples across different SNR scenarios. It is observed that the proportion of *case*-1 is exponentially increased with higher SNRs, which indicates that the proposed decoding-first mechanism can adaptively reduce the computational complexity under varying channel conditions. Fig. 12(b) further evaluates the error performance of detection-first (labeled "Det First") and decoding-first mechanism (labeled "proposed") in FCBsP IDD receivers. The results show that the proposed decoding-first IDD receiver significantly reduces the computational complexity of the traditional detection-first IDD receiver almost without performance degradation.

To evaluate the effectiveness of the proposed information compensation strategy, Fig. 13 compares the performance of the proposed FCBsP IDD receivers with and without the application of this strategy in a 32×12 LDPC-coded MIMO system with 64-QAM under high-rate ($R_c = 0.8$) and medium-rate ($R_c = 0.5$) scenarios. From the figure, the proposed FCBsP IDD receiver employing the information compensation strategy (labeled "with IC") delivers nearly 1.0 dB and 1.2 dB gains over its counterpart without information compensation (labeled "w/o IC") in the high- and medium-rate scenarios, respectively. The results show the effectiveness of the proposed information compensation strategy in mitigating potential performance degradation.

C. Performance Evaluation under practical Channels

In this subsection, we present the performance of the proposed FCBsP detector under practical P channels for both uncoded and coded MIMO systems.

1) Uncoded MIMO Scenarios: Fig. 14 provides the performance comparisons in a 32×12 64-QAM uncoded MIMO scenario with $\{16, 64, 256\}$ modulation schemes under practical channels. As illustrated, the CHEMP detector becomes increasingly difficult to converge as the modulation order



(a) 16-QAM (b) 64-QAM (c) 256-QAM

Fig. 14: BER performance of the proposed FCBsP detector in 32×12 uncoded MIMO scenarios with $\{16, 64, 256\}$-QAM modulation schemes under practical channels.

increases. In contrast, our proposed FCBsP detector enjoys faster convergence, with gains of approximately 0.5 dB and 2.2 dB gains over the BsP and LMMSE, respectively. Although the RD-GAI-BP demonstrates near-optimal error performance in the 16-QAM MIMO scenario, it suffers an SNR loss of nearly 0.8 dB compared to our proposed FCBsP detector in the high-modulated 256-QAM MIMO scenario.

2) Coded MIMO Scenarios: Fig. 15 demonstrates the error performance of our proposed FCBsP receivers in the same MIMO scenarios as Fig. 11, but under the more stringent practical channels. In the high-rate MIMO scenario (Fig. 15(a)), the CHEMP receivers fail to converge to the waterfall region. Compared to the LMMSE and BsP SDD receiver, our proposed FCBsP SDD receiver delivers 2.5 and 1.5 dB gains, respectively. Furthermore, our proposed FCBsP IDD receiver achieves a gain of 0.6 dB over the proposed FCBsP SDD receiver and 1.8 dB over the BsP IDD receiver, highlighting its resilience in challenging environments. In the medium-rate MIMO scenario (Fig. 15(b)), the CHEMP receivers converge in performance with the assistance of the channel coding. Our proposed FCBsP receivers still exhibit superior error performance. Specifically, our proposed FCBsP SDD receivers earn about 1.9 dB and 2.2 dB gains over the CHEMP and BsP SDD receivers, respectively. Similarly, the proposed FCBsP IDD receiver outperforms the CHEMP and BsP IDD receivers by about 0.8 dB and 1.9 dB gains. Overall, our proposed FCBsP receivers offer good performance efficiency and robustness under practical channel conditions.

VI. Conclusion

In this paper, a fixed-constellation version of the BsP (FCBsP) detector is proposed for MIMO turbo receivers. By truncating the constellation symbols to a fixed set and incorporating the approximate multi-user interference, the FCBsP detector offers both complexity and performance advantages over the BsP detector. To address the potential performance degradation caused by the missing counter-hypothesis issue in MIMO turbo systems, an information compensation strategy is introduced. Furthermore, a decoding-first mechanism is devised for the FCBsP-based MIMO turbo receiver, significantly reducing processing latency. Numerical results demonstrate that, compared to its counterparts, the proposed FCBsP detector contributes to lower complexity and better performance for MIMO turbo receivers under both Rayleigh and practical channels. Future work will focus on the hardware implementation and deployment of the proposed FCBsP MIMO turbo receivers in 6G CF-mMIMO systems.

Appendix A
Proof of the Equation (21)

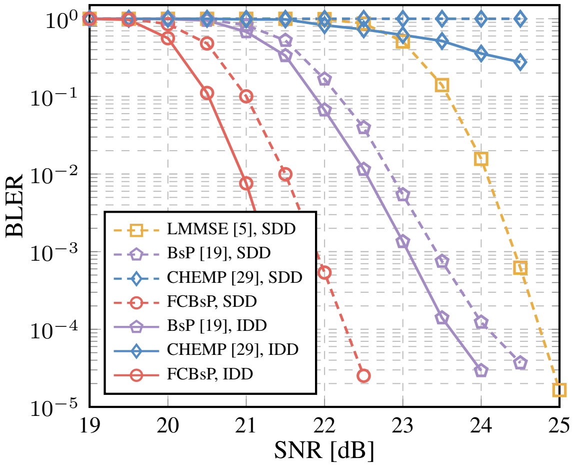
Take the soft messages sent from S_j to f_i as an example. The expression of the soft message $\alpha_{ji}(\varphi_k)$ is given as

$$\alpha_{ji}(\varphi_k) = \log \frac{P(s_j = \mu_{\varphi_k})}{P(s_j = \mu_1)},\ \mu_{\varphi_k} \in \mathcal{E}_j(n_m), \tag{31}$$

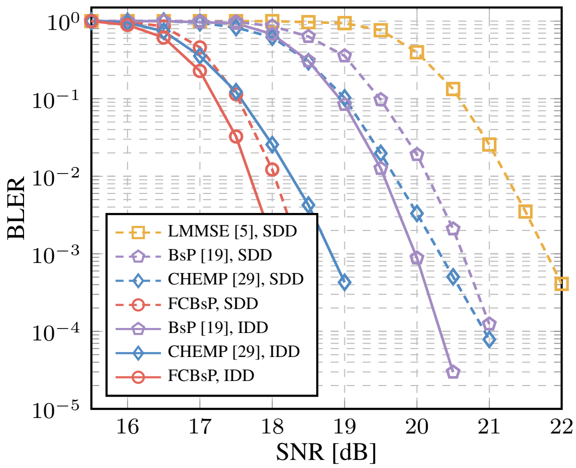
We only keep the n_m most reliable constellation symbols in each fixed configuration subset to compute the corresponding probability. It is worth noting that the reference constellation symbol μ_1 may not be involved in the fixed configuration subset, i.e., $\mu_1 \notin \mathcal{E}_j(n_m)$. To address this, we adjust the reference element to μ_{φ_1} in α expression. Using the property of logarithmic function, $P(s_j = \mu_1)$ can be replaced by $P(s_j = \mu_{\varphi_1})$ as

$$\begin{aligned}\alpha_{ji}(\varphi_k) - \alpha_{ji}(\varphi_1) &= \log \frac{P(s_j = \mu_{\varphi_k})}{P(s_j = \mu_1)} - \log \frac{P(s_j = \mu_{\varphi_1})}{P(s_j = \mu_1)} \\ &= \log \frac{P(s_j = \mu_{\varphi_k})}{P(s_j = \mu_{\varphi_1})}.\end{aligned} \tag{32}$$

Because only n_m reliable constellation symbols are considered for probability computation, it is reasonable to assume the



(a) $(N,\ K)$ = (1440, 1152)



(b) $(N,\ K)$ = (1440, 720)

Fig. 15: BLER performance of the proposed FCBsP-based receiver in 32×12 64-QAM LDPC-coded MIMO systems under practical channels.

probability for the remaining $|\mathcal{Q}| - n_m$ constellation symbols is zero. In this way, the probability of μ_{φ_1} is derived as

$$\begin{aligned}&\sum_{k=1}^{n_m} \exp\left(\alpha_{ji}(\mu_{\varphi_k}) - \alpha_{ji}(\varphi_1)\right)\\ =&\frac{P(s_j = \mu_{\varphi_1}) + P(s_j = \mu_{\varphi_2}) + \cdots + P(s_j = \mu_{\varphi_{n_m}})}{P(s_j = \mu_{\varphi_1})} \\ =&\frac{1}{P(s_j = \mu_{\varphi_1})}.\end{aligned} \tag{33}$$

Thus, $P(s_j = \mu_{\varphi_1})$ can be expressed as

$$P(s_j = \mu_{\varphi_1}) = \frac{1}{\sum\limits_{k=1}^{n_m} \exp\left(\alpha_{ji}(\mu_{\varphi_k}) - \alpha_{ji}(\varphi_1)\right)}. \tag{34}$$

Based on the Eq. (34) and Eq. (32), the probabilities of other constellation symbols are given as

$$P(s_j = \mu_{\varphi_k}) = \frac{\exp\left(\alpha_{ji}(\varphi_k) - \alpha_{ji}(\varphi_1)\right)}{\sum_{k=1}^{n_m} \exp\left(\alpha_{ji}(\mu_{\varphi_k}) - \alpha_{ji}(\varphi_1)\right)}. \tag{35}$$

To reduce the computational complexity, we apply the *max-log Approximation* operation to Eq. (35). Therefore, Eq. (35) can be simplified as

$$P(s_j = \mu_{\varphi_k}) \approx \exp\left(\alpha_{ji}(\varphi_k) - \max_{\mu_{\varphi_m} \in \mathcal{E}_j(n_m)} \alpha_{ji}(\varphi_m)\right). \tag{36}$$

We note that this approximation requires probability normalization to ensure the probabilities sum to one.

References

[1] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive MIMO for next generation wireless systems," *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 186–195, 2014.

[2] X. You, C.-X. Wang, J. Huang, X. Gao, Z. Zhang, M. Wang, Y. Huang, C. Zhang, Y. Jiang, J. Wang *et al.*, "Towards 6G wireless communication networks: Vision, enabling technologies, and new paradigm shifts," *Sci. China Inf. Sci.*, vol. 64, pp. 1–74, 2021.

[3] S. Yang and L. Hanzo, "Fifty years of MIMO detection: The road to large-scale MIMOs," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 1941–1988, 2015.

[4] M. A. Albreem, M. Juntti, and S. Shahabuddin, "Massive MIMO detection techniques: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3109–3132, 2019.

[5] N. Kim, Y. Lee, and H. Park, "Performance analysis of MIMO system with linear MMSE receiver," *IEEE Trans. Wireless Commun.*, vol. 7, no. 11, pp. 4474–4478, 2008.

[6] G. Peng, L. Liu, S. Zhou, S. Yin, and S. Wei, "A 1.58 Gbps/W 0.40 Gbps/mm^2 ASIC implementation of MMSE detection for 128×8 64-QAM massive MIMO in 65 nm CMOS," *IEEE Trans. Circuits Syst. I*, vol. 65, no. 5, pp. 1717–1730, 2017.

[7] *Study on enhancement of Ultra-Reliable Low-Latency Communication (URLLC) support in the 5G core network*, 3GPP TR 23.725, Jun. 2019.

[8] J. Cespedes, P. M. Olmos, M. Sánchez-Fernández, and F. Perez-Cruz, "Expectation propagation detection for high-order high-dimensional MIMO systems," *IEEE Trans. Commun.*, vol. 62, no. 8, pp. 2840–2849, 2014.

[9] G. Yao, H. Chen, and J. Hu, "An improved expectation propagation based detection scheme for MIMO systems," *IEEE Trans. Commun.*, vol. 69, no. 4, pp. 2163–2175, 2021.

[10] X. Tan, M. Ding, Z. Ji, Z. Zhang, X. You, and C. Zhang, "An improved power expectation propagation detector for massive MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 1353–1357, 2024.

[11] C. Jeon, R. Ghods, A. Maleki, and C. Studer, "Optimality of large MIMO detection via approximate message passing," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2015, pp. 1227–1231.

[12] J. Ma and L. Ping, "Orthogonal AMP," *IEEE Access*, vol. 5, pp. 2020–2033, 2017.

[13] L. Liu, S. Huang, and B. M. Kurkoski, "Memory AMP," *IEEE Trans. Inf. Theory*, vol. 68, no. 12, pp. 8015–8039, 2022.

[14] J. Hu and T. M. Duman, "Graph-based detection algorithms for layered space-time architectures," *IEEE J. Sel. Areas Commun.*, vol. 26, no. 2, pp. 269–280, 2008.

[15] P. Som, T. Datta, A. Chockalingam, and B. S. Rajan, "Improved large-MIMO detection based on damped belief propagation," in *Proc. IEEE Inf. Theory Works. (ITW)*, 2010, pp. 1–5.

[16] J. Goldberger and A. Leshem, "MIMO detection for high-order QAM based on a gaussian tree approximation," *IEEE Trans. Inf. Theory*, vol. 57, no. 8, pp. 4973–4982, 2011.

[17] J. Yang, W. Song, S. Zhang, X. You, and C. Zhang, "Low-complexity belief propagation detection for correlated large-scale MIMO systems," *J. Signal Process. Syst.*, vol. 90, no. 4, pp. 585–599, 2018.

[18] Y. Zhang, W. Zhou, Y. Zhang, H. Ji, Y. Huang, X. You, and C. Zhang, "2.7 BayesBB: A 9.6 Gbps 1.61 ms configurable all-messagepassing baseband-accelerator for B5G/6G cell-free massive-MIMO in 40nm CMOS," *IEEE ISSCC Dig. Tech. Papers*, vol. 67, pp. 48–50, Feb. 2024.

[19] W. Zhou, Y. Shen, L. Li, Y. Huang, C. Zhang, and X. You, "Belief-selective propagation detection for MIMO systems," *IEEE Trans. Commun.*, vol. 71, no. 12, pp. 7244–7257, 2023.

[20] W. Zhou, Z. Ji, Z. Tan, Z. You, X. Tan, X. You, and C. Zhang, "Approximate belief-selective propagation detector for massive MIMO systems," *IEEE Trans. Circuits Syst. I*, vol. 71, no. 6, pp. 2938–2950, 2024.

[21] Z. Tan, W. Zhou, Y. Huang, C. Zhang, and X. You, “TrellisBsP: Trellis-based belief-selective propagation for MIMO detection,” *IEEE Wireless Commun. Lett.*, 2025, Early Access.

[22] B. M. Hochwald and S. Ten Brink, “Achieving near-capacity on a multiple-antenna channel,” *IEEE Trans. Commun.*, vol. 51, no. 3, pp. 389–399, 2003.

[23] X. Bian, Y. Mao, and J. Zhang, “Joint activity detection, channel estimation, and data decoding for grant-free massive random access,” *IEEE Internet Things J.*, vol. 10, no. 16, pp. 14 042–14 057, 2023.

[24] R. B. Di Renna and R. C. de Lamare, “Joint channel estimation, activity detection and data decoding based on dynamic message-scheduling strategies for mMTC,” *IEEE Trans. Commun.*, vol. 70, no. 4, pp. 2464–2479, 2022.

[25] X. Tan, W. Li, Z. Zhang, X. You, and C. Zhang, “An efficient approximate EP based iterative detection and decoding for massive MIMO,” *IEEE Wireless Commun. Lett.*, vol. 13, no. 2, pp. 255–259, 2024.

[26] X. Tan, Q. Qian, Y. Cai, Z. Zhang, Y. Huang, X. You, and C. Zhang, “Expectation propagation-based parallel iterative detection and decoding for massive MIMO,” *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 93–106, 2024.

[27] X. Zhou, J. Zhang, C.-K. Wen, S. Jin, and S. Han, “Graph neural network-enhanced expectation propagation algorithm for MIMO turbo receivers,” *IEEE Trans. Signal Process.*, vol. 71, pp. 3458–3473, 2023.

[28] M. Nassar, P. Schniter, and B. L. Evans, “A factor graph approach to joint OFDM channel estimation and decoding in impulsive noise environments,” *IEEE Trans. Signal Process.*, vol. 62, no. 6, pp. 1576–1589, 2013.

[29] T. L. Narasimhan and A. Chockalingam, “Channel hardening-exploiting message passing (CHEMP) receiver in large-scale MIMO systems,” *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 847–860, 2014.

[30] J. Qian, S. Hu, and H. Wang, “Joint MIMO detection and LDPC decoding via enhanced belief propagation for 5G-NR,” in *Proc. IEEE Wire. Commun. Net. Conf. (WCNC)*, 2022, pp. 1093–1098.

[31] J. Zhang, H. Wang, J. Qian, and Z. Gao, “Soft MIMO detection using marginal posterior probability statistics,” in *Proc. IEEE GLOBECOM*, 2022, pp. 3198–3204.

[32] F. Long, T. Lv, R. Cao, and H. Gao, “Single edge based belief propagation algorithms for MIMO detection,” in *Proc. IEEE Sarnoff Symp.*, 2011, pp. 1–5.

[33] J. Chen and M. P. Fossorier, “Density evolution for two improved bp-based decoding algorithms of LDPC codes,” *IEEE Commun. Lett.*, vol. 6, no. 5, pp. 208–210, 2002.

[34] R. Wang and G. Giannakis, “Approaching MIMO channel capacity with reduced-complexity soft sphere decoding,” in *Proc. IEEE Wire. Commun. Net. Conf. (WCNC)*, vol. 3, 2004, pp. 1620–1625 Vol.3.

[35] S. Chen, T. Zhang, and Y. Xin, “Relaxed K-best MIMO signal detector design and VLSI implementation,” *IEEE Trans. VLSI Syst.*, vol. 15, no. 3, pp. 328–337, 2007.

[36] A. Voicila, D. Declercq, F. Verdier, M. Fossorier, and P. Urard, “Low-complexity decoding for non-binary LDPC codes in high order fields,” *IEEE Trans. Commun.*, vol. 58, no. 5, pp. 1365–1375, 2010.

[37] D. Wang, X. You, Y. Huang, W. Xu, J. Li, P. Zhu, Y. Jiang, Y. Cao, X. Xia, Z. Zhang *et al.*, “Full-spectrum cell-free RAN for 6G systems: System design and experimental results,” *Sci. China Inf. Sci.*, vol. 66, no. 3, pp. 130 305:1–130 305:14, 2023.

[38] 3GPP, “NR; Physical layer procedures for data,” 3rd Generation Partnership Project (3GPP), Technical Specification (TS) 38.214, 01 2018, version 15.0.0. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.214/

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